Predicting Customer Response to Marketing Campaigns

Retail Analytics Capstone Project

Marina Ferreira

The Business Question

?

A retail company sends marketing campaigns to its customers.

Only some customers respond.

Can we predict, in advance, which customers are likely to respond to the next campaign?

If we can, the company can target the right customers
and get a better return on every campaign it sends.

The Data

Customer records are from a retail loyalty program, covering demographics, purchase history, and past marketing campaigns.

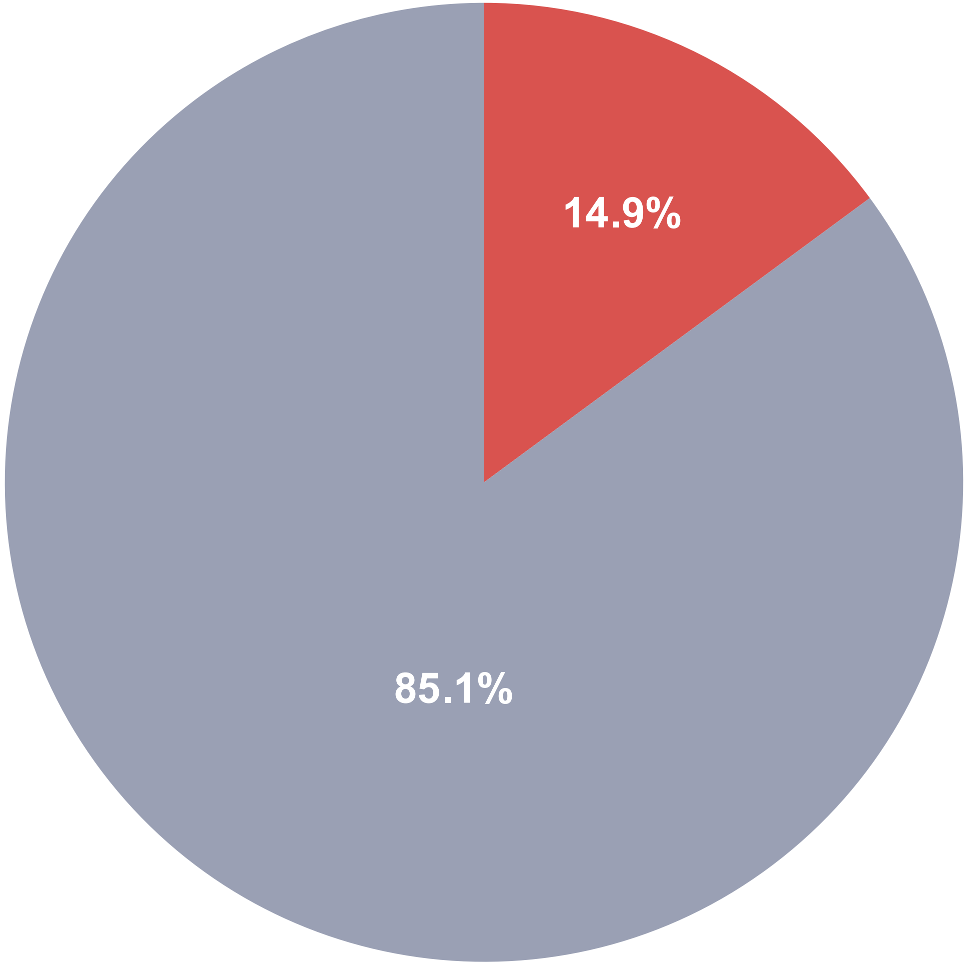
2,240
Customers

27
Variables tracked

6
Past marketing campaigns

Demographics • Spending by product category • Purchase channels • Campaign history

Response Is Rare



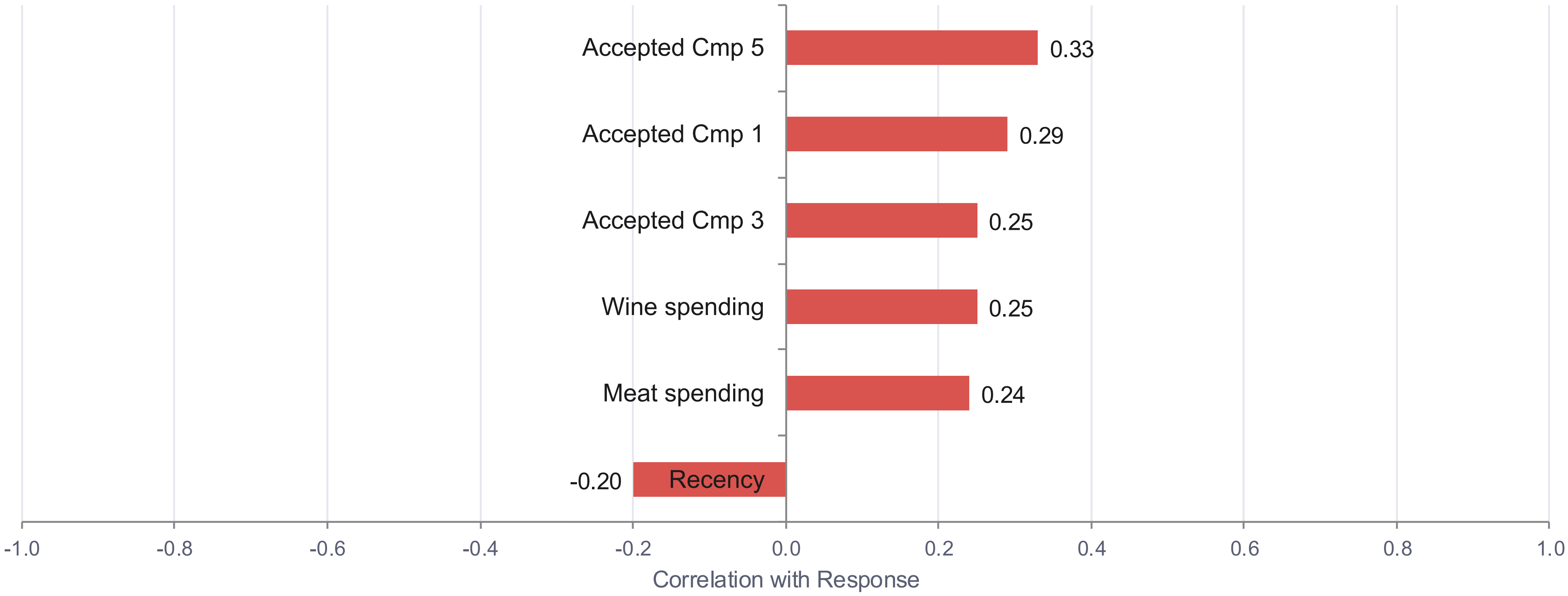
14.9%

of customers responded
to the last campaign

The challenge isn't reaching customers — it's knowing which ones are actually likely to say yes.

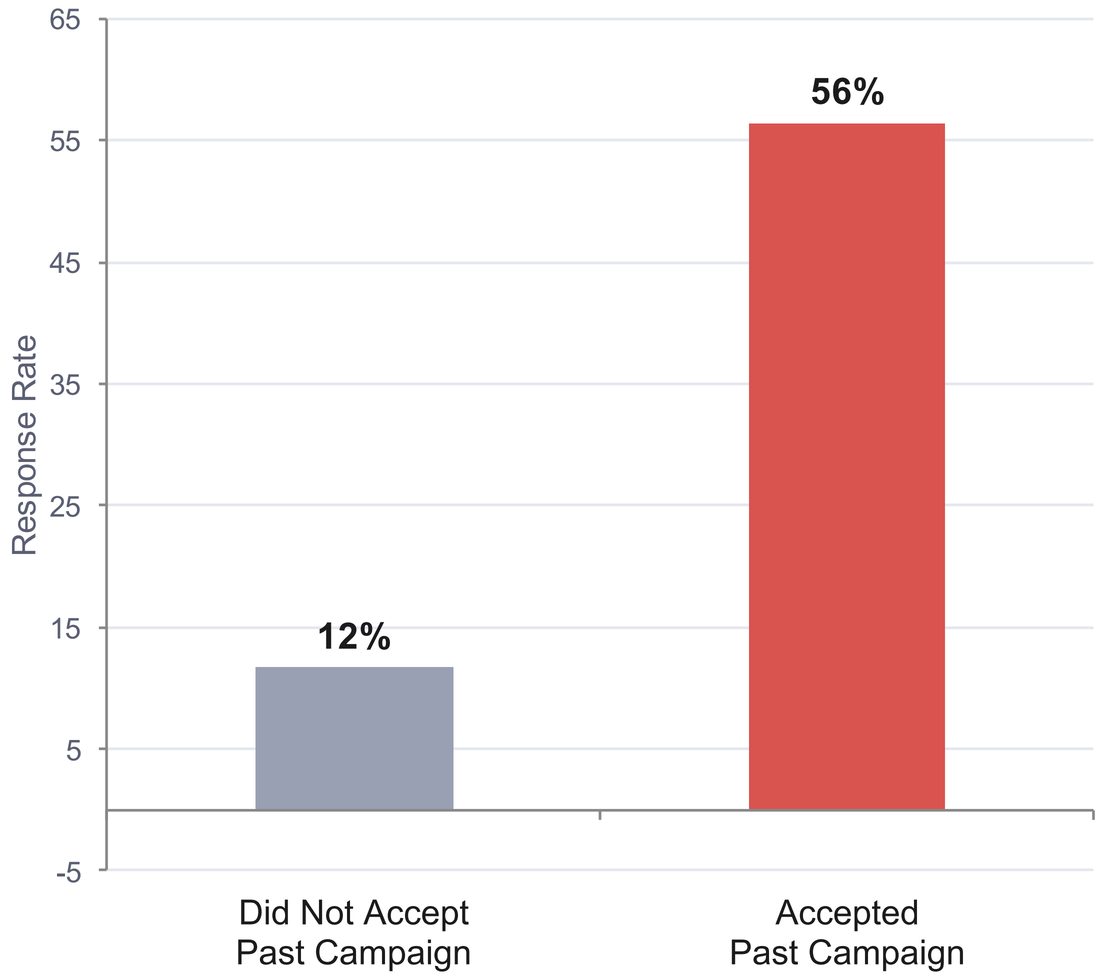
What Predicts a Response?

Correlation of each variable with campaign response (scale: -1 to 1)



Past campaign behavior and spending matter far more than age, income, or household size.

Past Behavior Is a Strong Signal



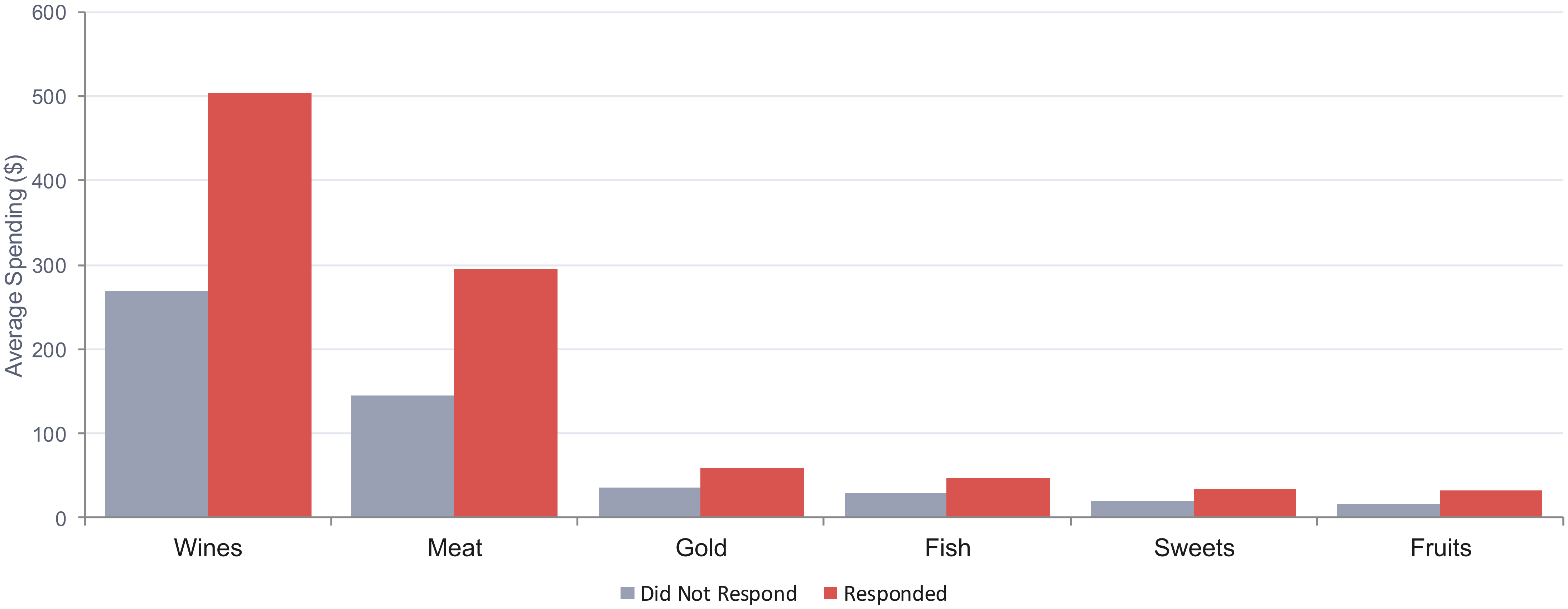
~5x

more likely to respond if they accepted a past campaign

This pattern held consistently across the three campaigns with the strongest correlation to Response.

Responders Spend More

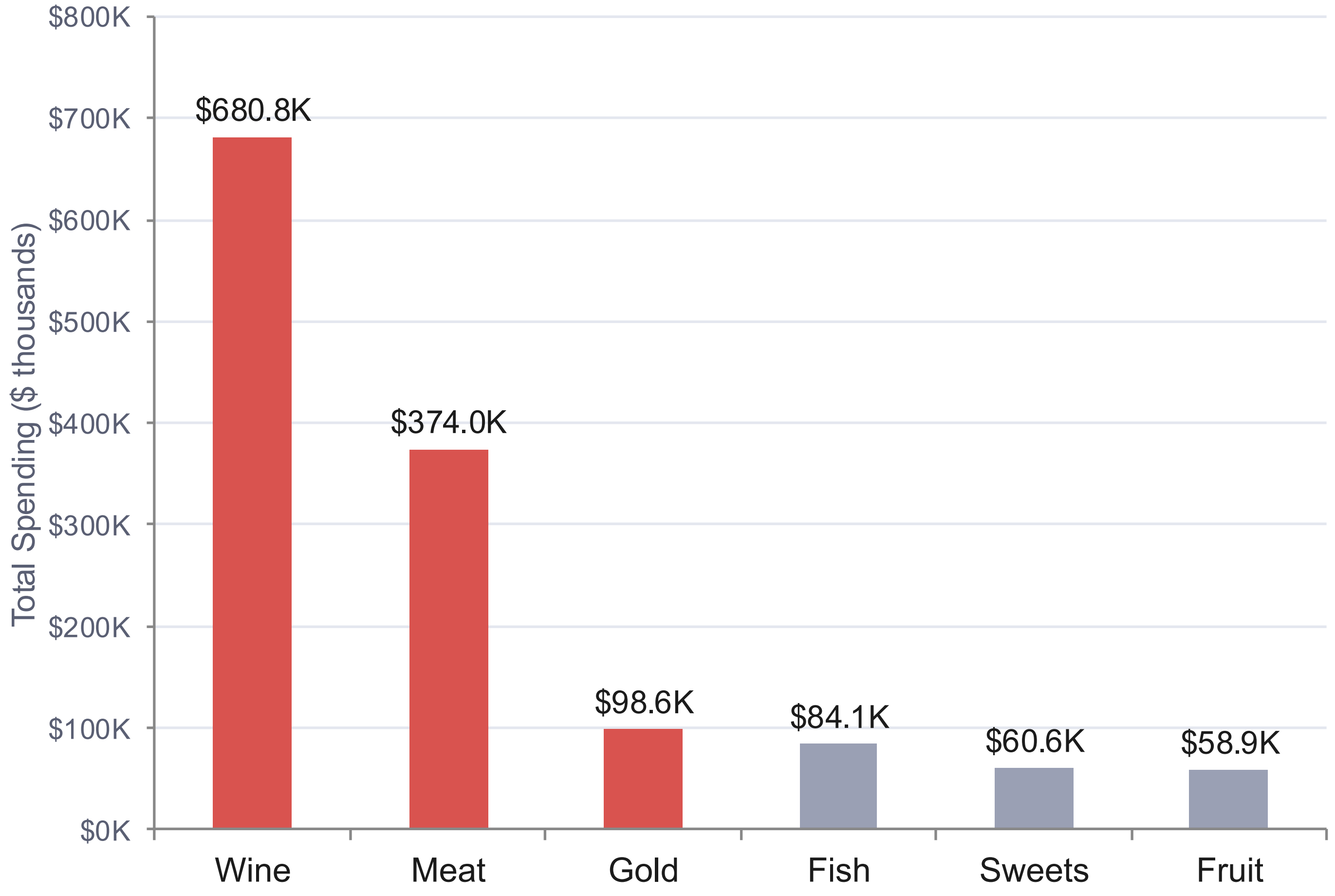
Average spending per customer, by product category



The gap is widest for wine and meat — the same two categories that showed up among the top predictors of response.

Spending Is Concentrated

Total spending by product category, high to low



85%

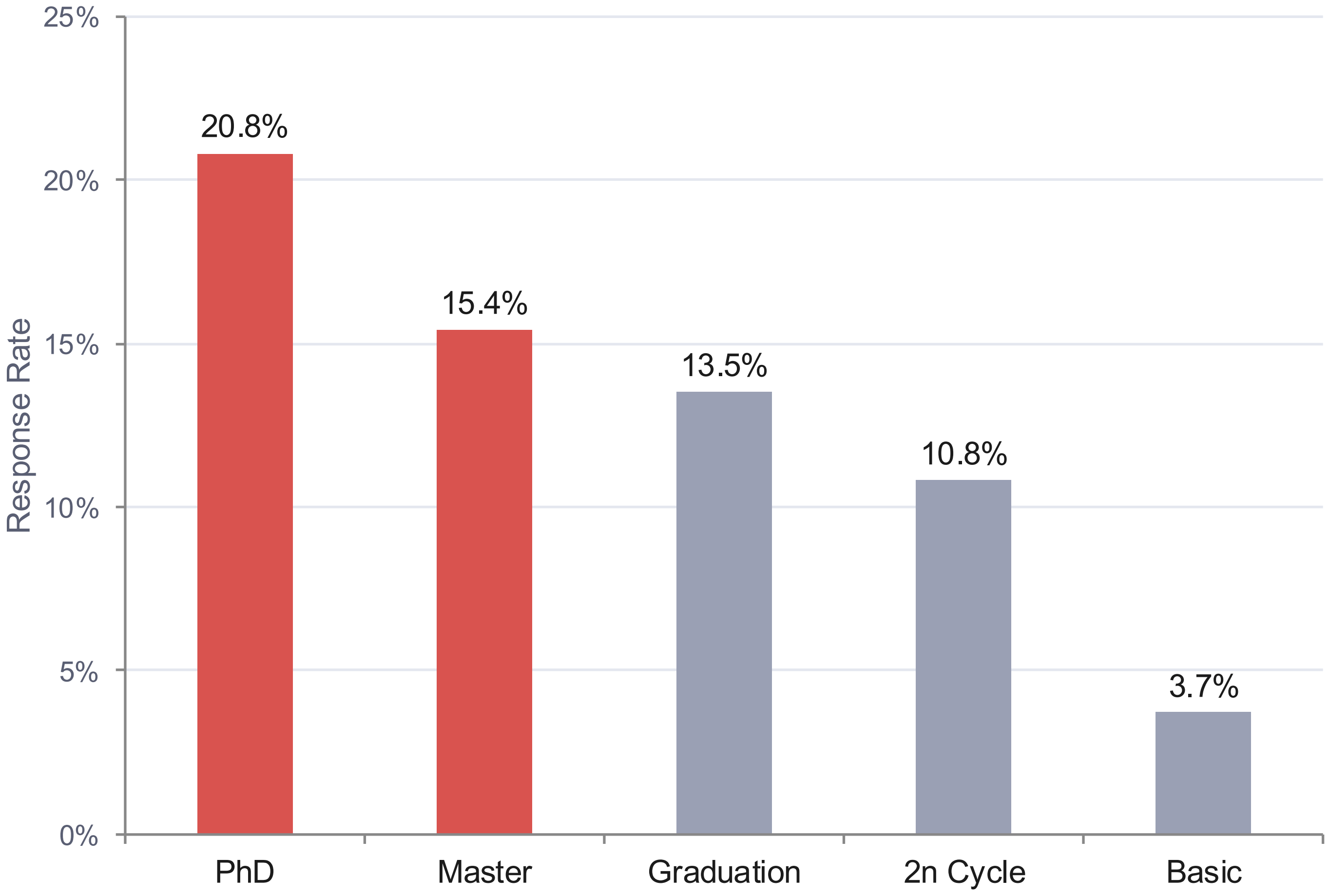
of all spending comes from Wine, Meat & Gold

Wine alone accounts for about half of everything customers spend across all categories.

A Pareto view: a small number of categories drive most of the spending.

Education Also Predicts Response

Response rate by education level (Chi-Square test: p = 0.0001)



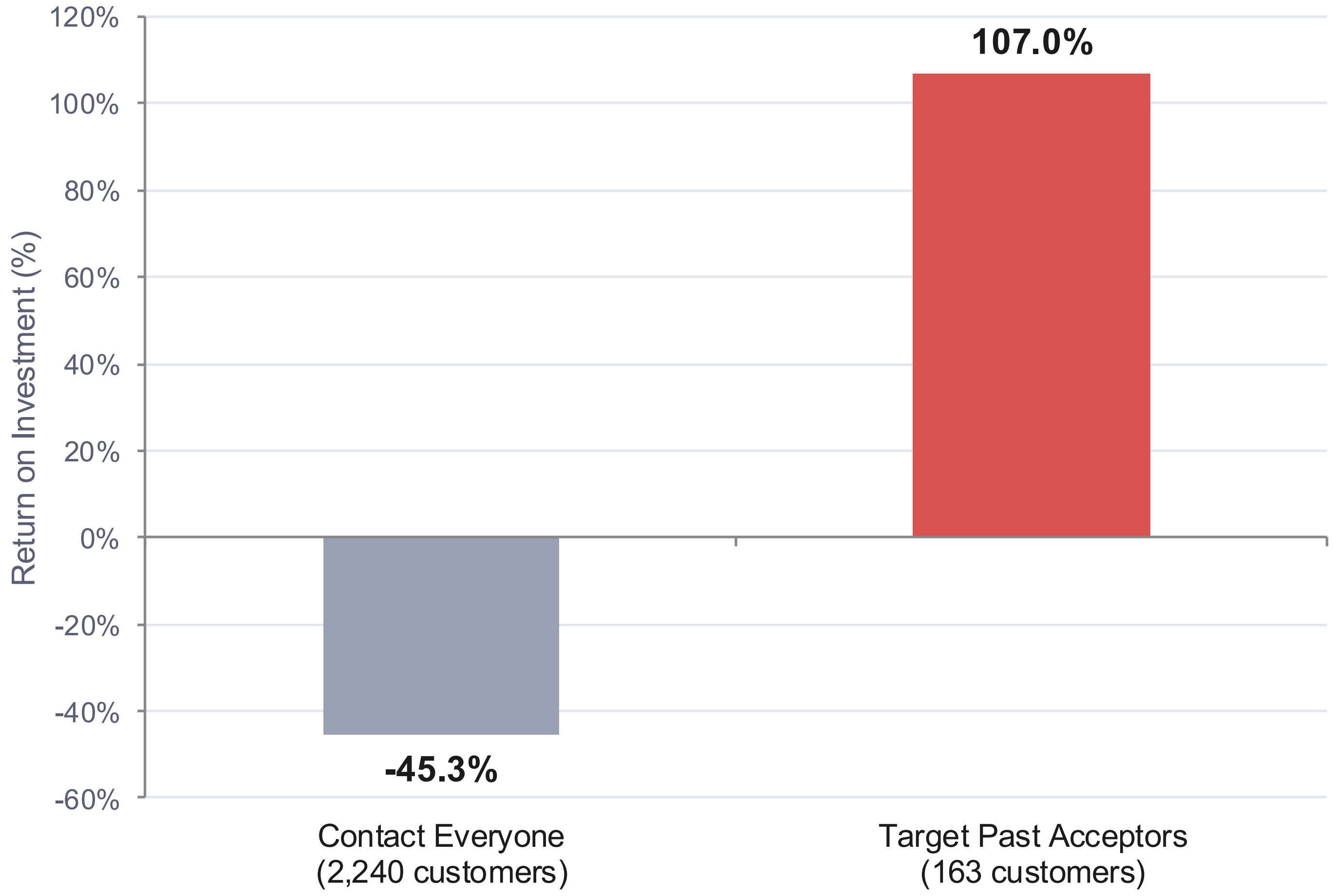
~5.6x

more likely to respond: PhD vs. Basic education

This didn't show up in the correlation matrix because Education is categorical — so we used a Chi-Square test instead.

From Loss to Profit

Return on investment (ROI): profit earned per dollar spent contacting customers



+107%

ROI when targeting past acceptors — more than double the money spent

Mailing every customer returns -45% — every dollar spent loses money overall. Targeting past acceptors flips that into a +107% return instead.

ROI = [(responses × $11 revenue) − (customers contacted × $3 cost)] ÷ (customers contacted × $3 cost)

Recommendation

1. **Target customers with prior campaign acceptance**

   They respond at roughly 5x the rate of non-acceptors — enough to turn the campaign profitable.

2. **Prioritize high spenders in wine, meat & gold**

   These 3 categories make up about 85% of all customer spending (Pareto analysis).

3. **Segment campaigns by education level**

   PhD and Master's customers respond up to ~5.6x more than Basic — confirmed with a Chi-Square test.

4. **Combine signals, don't rely on one factor**

   No single variable is a strong predictor alone — a model blending several works best.